



2. (continued)

(c) Show by calculation whether the collision is elastic or inelastic.

4

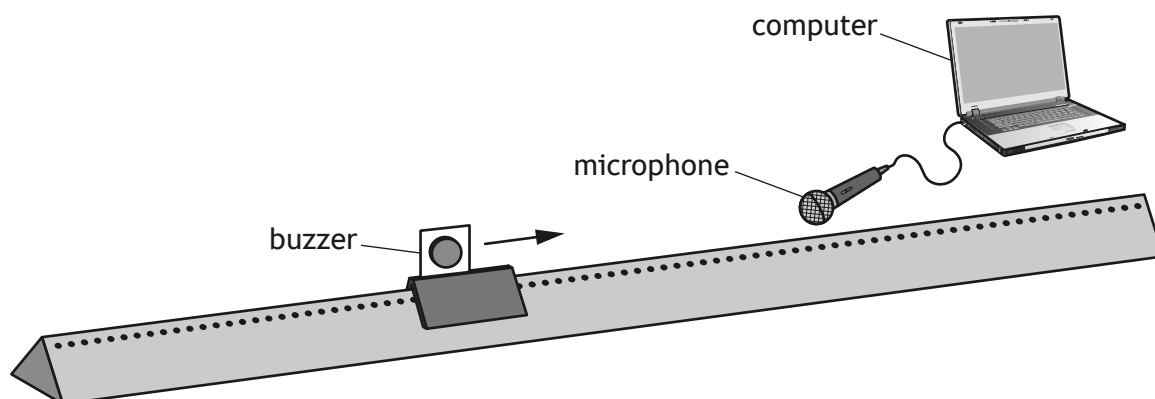
*Space for working and answer*

[Turn over



2. (continued)

- (d) The technician now sets up the linear air track to demonstrate the Doppler effect.



The vehicle is fitted with a buzzer. A microphone is connected to a computer. The vehicle is held stationary and the buzzer is switched on. The buzzer emits sound with a frequency of 845 Hz.

The vehicle now travels at a constant speed towards the microphone. The frequency of sound measured by the computer is 849 Hz.

- (i) Calculate the speed of the vehicle along the linear air track.

*Space for working and answer*

3



2. (d) (continued)

- (ii) Explain, in terms of wavefronts, why the frequency of the sound detected by the microphone is greater than the frequency of the sound emitted by the buzzer.

You may wish to use a diagram.

1

- (iii) A student standing at the end of the linear air track, near the microphone, states that they did not hear this small change in frequency. Using the same apparatus, describe one way the Doppler effect could be made more obvious for the student.

1

[Turn over

